

Common-reference workflow for runoff-extreme frequency attribution

1 Inputs

Daily qtot
7 GHMs × 3 scenarios
1950–2019; native calendars

2 Common SRI reference

30-day backward mean
Fit 12 monthly distributions
to factual obsclim-histsoc
per model and grid cell

3 Events within models

Use the same 12 fits
for all three scenarios
Detect events separately in
each model–scenario
No model-mean SRI

4 Frequency change

Drought: $SRI \leq -0.5$; ≥ 20 d
Flood: $SRI \geq 0.5$; reset 20 d
P1: 1950–1984; P2: 1985–2019
 $FR = (N_2 + 0.5)/(N_1 + 0.5)$

5 Paired attribution

$\Delta HA = FR_{hist} - FR_{1901}$
 $\Delta CC = FR_{obs} - FR_{hist}$
 $\Delta Total = \Delta CC + \Delta HA$
Basin DIndex from $|\Delta|$

QUALITY CONTROL AND ROBUSTNESS

QC

Exclude failed monthly fits
or internal SRI gaps
retain every grid-level QC row

Calibration

Primary: factual common reference
Sensitivity: scenario-specific fits
compare grid and basin results

Definitions

Thresholds: 0.5, 0.8, 1.0
duration/reset: 10, 20, 30 d
split years: 1980, 1985, 1990

Aggregation

paired contrasts within model
 $\cos(\text{latitude})$ basin weighting
equal model weights and LOMO